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## Bagian I

Tentukan amplitudo, periode dan gambarkan grafik fungsi trigonometri (manual dan bantuan alat) berikut untuk satu periode!

1). $y = 2 \sin x$ 2). $y = 3 \sin x$ 3). $y = 4 \sin x$

## Bagian II

Gunakan prinsip pergeseran dari bentuk $y1 = R \sin x$ atau $y2 = R \cos x$ untuk grafik fungsi trigonometri untuk satu periode berikut:

1). $y = 3 \sin (x + \frac{\pi}{3})$ 3) $y = 4 \cos (x + \frac{\pi}{4})$

2). $y = 3 \sin (x - \frac{\pi}{3})$

## Bagian III

Gunakan prinsip pergeseran dari bentuk $y1 = R \sin \omega t$ ~~dan~~ atau $Y2 = R \cos \omega t$ untuk grafik fungsi trigonometri untuk satu periode, dan berapa amplitudo dan periode (dalam detik) dengan $\omega$ (kecepatan sudut) serta lengkapi grafiknya berikut:

1) $y = 3 \sin (2t + \frac{\pi}{3})$ 3) $y = 4 \cos (3t + \frac{\pi}{4})$

2) $y = 3 \sin (2t - \frac{\pi}{3})$

## Bagian III

Tentukan amplitudo, waktu periodik, frekuensi dan sudut phase (leading / lagging $R \sin \omega t$) dan grafik

1). $y = 30 \sin (50\pi t + 0{,}36)$ 3) $i = 15 \sin (314{,}2t + 0{,}468)$

2). $x = 22{,}4 \sin (40t - 0{,}60)$

## Bagian IV

1). Arus bolak balik rangkaian listrik memenuhi persamaan $i(t) = 32 \sin (3\pi t)$ dengan $t \geq 0$. hitung:

a) arus pada $t = 0{,}5$ ms dan $t = 2$ ms

b) waktu t pada arus maksimal

c) waktu t pada arus $i = 16$ kedua

2). tegangan bolak balik rangkaian listrik pada setiap t diberikan persamaan $V = 70 \sin (100\pi t + 0{,}380)$. hitunglah

a) tegangan pada $t = 0$ dan $t = 5$ ms

b) waktu t pada saat tegangan mencapai maksimum

c) sketsa grafik osilasi tegangan untuk satu putaran.

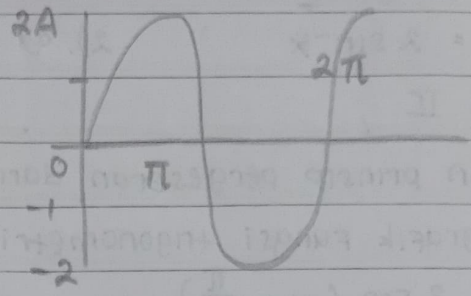
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# Jawaban

## Bagian I

1. amplitudo = 2

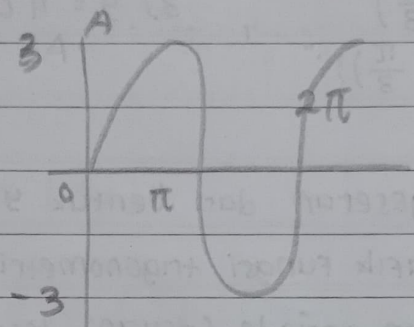
periode = $2\pi$




2. $y = 3 \sin x$

amplitudo = 3

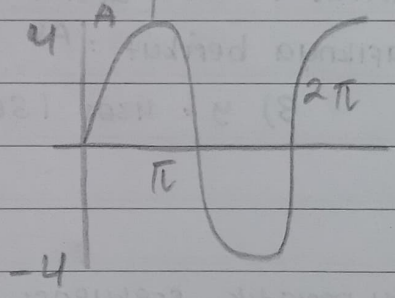
periode = $2\pi$




3. $y = 4 \sin x$

amplitudo = 4

periode = $2\pi$



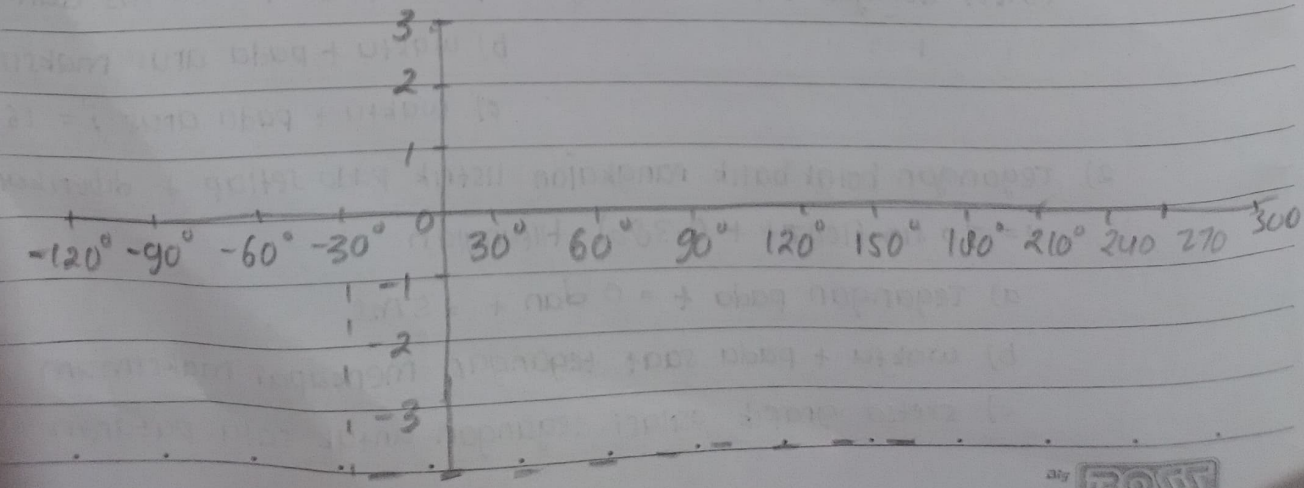

## Bagian II

1. $y = 3 \sin \left(x + \frac{\pi}{3}\right)$

Amplitudo = 3

periode = $2\pi$

pergeseran = ke kiri $\frac{\pi}{3} = 60°$ kiri

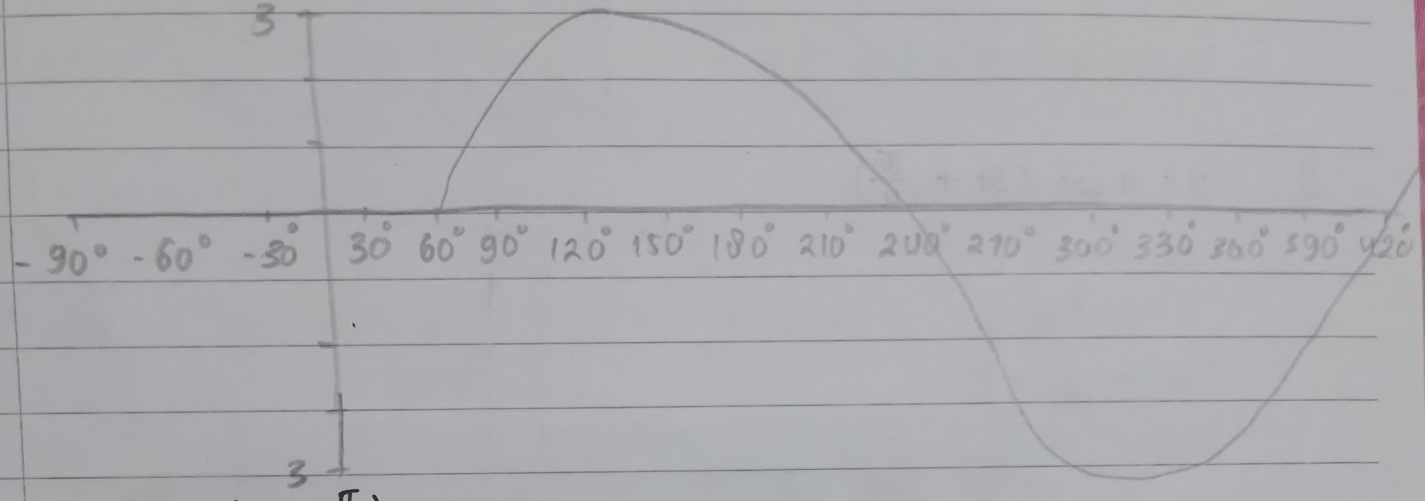


2. $y = 3 \sin (x - \frac{\pi}{3})$

amplitudo = 3

periode = $2\pi$

pergeseran = $\frac{\pi}{3} = 60^\circ$ ke kanan

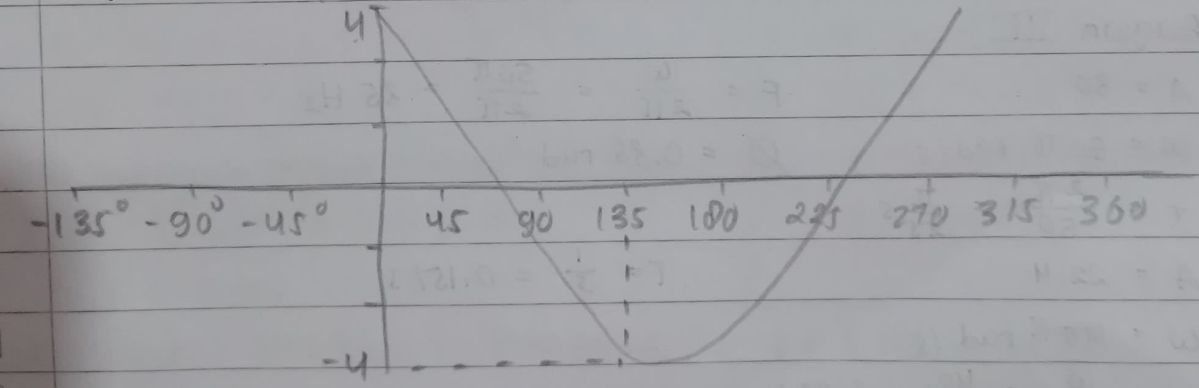



3. $y = 4 \cos (x + \frac{\pi}{4})$

amplitudo = 4

periode = $2\pi$

pergeseran = $\frac{\pi}{4} = 45^\circ$ (ke kiri)




Bagian II

1. $3 \sin (2t + \frac{\pi}{3}) = y$

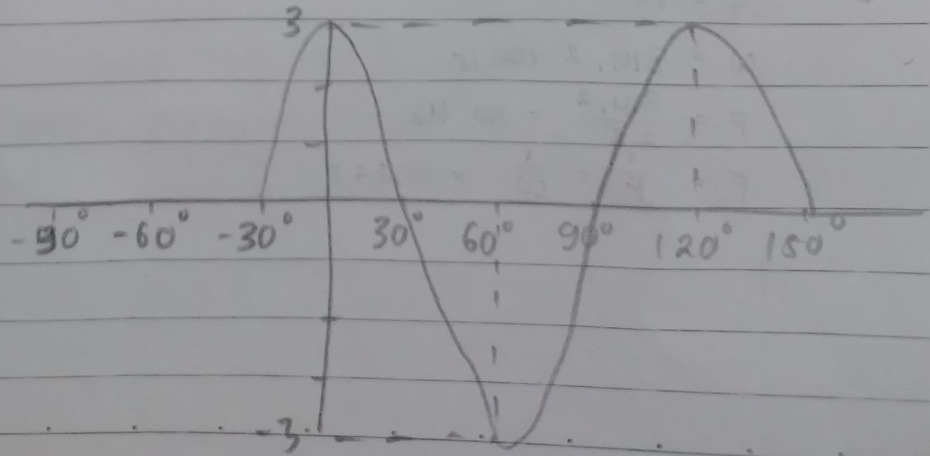
$A = 3$

$T = \frac{2\pi}{2} = \pi$

$2t + \frac{\pi}{3} = 0$

$2t = -\frac{\pi}{3}$

$t = -\frac{\pi}{6} = -30^\circ$



2. $y = 3 \sin (2t - \frac{\pi}{3})$

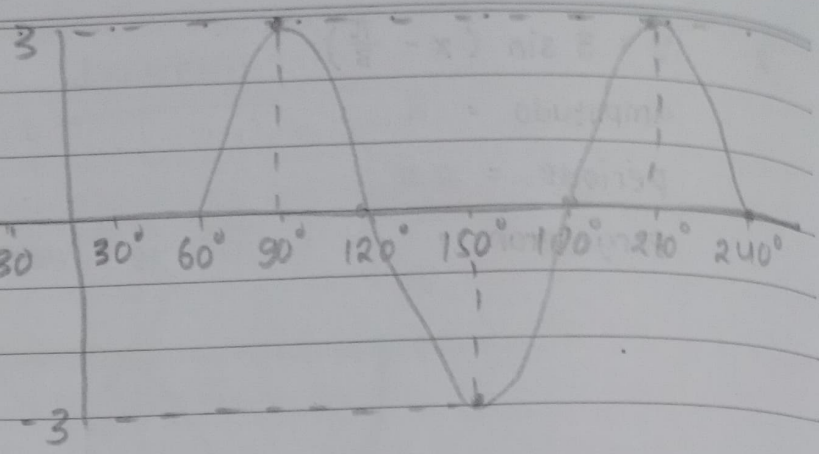
$A = 3$

$T = \frac{2\pi}{\pi} = \pi$

$2t = \frac{\pi}{3}$

$t = \frac{\pi}{6}$

geser kanan $= 60°$




3. $y = 4 \cos (3t + \frac{\pi}{4})$

$A = 4$

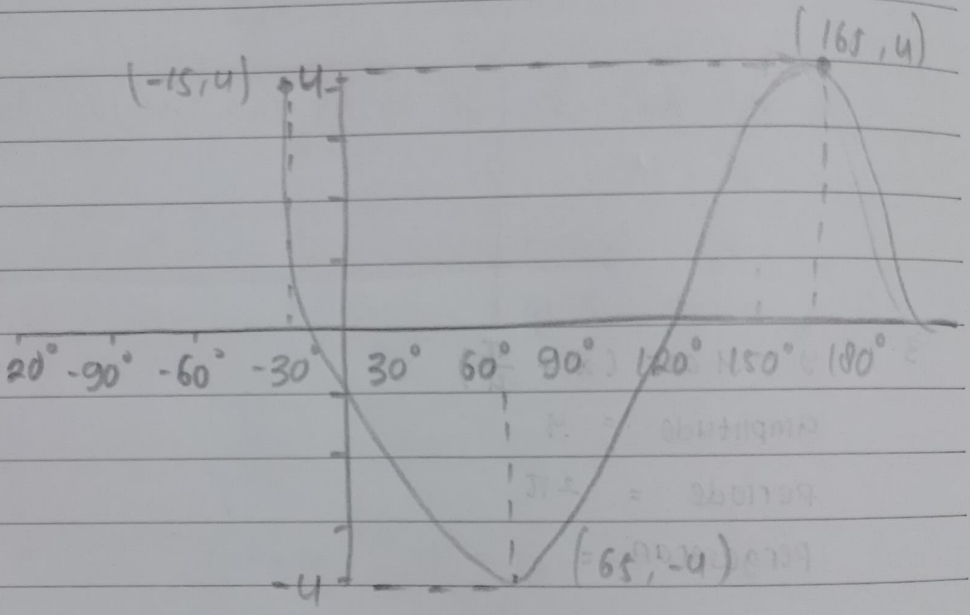
$T = \frac{2\pi}{3}$

$3t + \frac{\pi}{4} = 0$

$3t = -\frac{\pi}{4}$

$t = -\frac{\pi}{12} = -15°$

geser kiri $= 45°$




Bagian III

1. $A = 30$

$\omega = 50\pi$ rad/s

$T = \frac{2\pi}{50} = \frac{\pi}{25}$ s

$F = \frac{\omega}{2\pi} = \frac{50\pi}{2\pi} = 25$ Hz

$\varnothing = 0,36$ rad (positif leading terhadap sin (wt)

2. $A = 22,4$

$\omega = 40\pi$ rad/s

$s = \frac{\omega}{2\pi} = \frac{40}{2\pi} = 6,37$ Hz

$T = \frac{1}{5} = 0,157$ s

Sudut pase $= -0,60$ rad (negatif, sehingga leading terhadap sin wt)

3. $A = 15$

$\omega = 314,2$ rad/s

$F = \frac{314,2}{2\pi} = 50$ Hz

$T = \frac{1}{F} = \frac{1}{50} = 0,02$ s

Sudut pase $= 0,0468$ rad (positif, sehingga leading terhadap (wt))

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Bagian IV

1. a arus pada $t = 0{,}5$ ms dan $t = 2$ ms

$t = 0{,}5$ ms

$t = 0{,}5$ ms

• $t = 0{,}5$ ms

$i(0{,}5) = 32 \sin(3\pi \cdot 0{,}5)$

$= 32 \sin(1{,}5\pi \text{ rad})$

$\sim \sin(1{,}5\pi \text{ rad}) = -1$

$i(0{,}5) = 32 \cdot (-1) = -32$ A

$t = 2$ ms

$i(2) = 32 \sin(3\pi \cdot 2)$

$= 32 \sin(6\pi)$

$\sim \sin(6\pi \text{ rad}) = 0$

$i(2) = 32 \cdot 0 = 0$ A

b waktu t pada saat arus maksimum pertama

arus i maks (A) = 32 A

$32 = 32 \sin(3\pi t)$

$= \sin(3\pi t)$

$3\pi t = \frac{\pi}{2}$ rad

$t = \frac{1}{6}$ det $= 0{,}1667$ det $= 166{,}7$ ms

c waktu t pada saat arus i = i6 kedua

$16 = 32 \sin(3\pi t)$ $(n = 0)$

$\sin(3\pi t) = \frac{1}{2}$

$3\pi t = \frac{\pi}{6}$ rad

$t = \frac{1}{18}$ det $= 0{,}0556$ det

$t = 55{,}56$ ms

$16 = 32 \sin(3\pi t)$ $(n = 1)$

$\sin(3\pi t) = \frac{1}{2}$

$3\pi t = 2\pi$ rad

$t = \frac{2}{3}$ det $= 0{,}66667$ det

$t = 666{,}67$ ms

2. a $t = 0$

$v(0) = 70 \sin(100\pi - 0t + 0{,}380)$

$= 70 \sin(0{,}380)$

$v = 70 \sin 21° 46'$

$v \sin(0{,}380) = 0{,}3709$

$v(0) = 70 \times 0{,}3909$

$= 25{,}96$ V

$t = 5$ ms $\rightarrow 0{,}005$ s

$v(0{,}005) = 70 \sin(100\pi \cdot 0{,}0005 + 0{,}380)$

$= 70 \sin(\frac{\pi}{2} + 0{,}380)$

$\sin(\frac{\pi}{2} + 0{,}380) \rightarrow \cos(0{,}380)$

$\cos(0{,}380) = 0{,}928$

$v(0{,}005) = 70 \times 0{,}928 = 64{,}96$ V

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b $V_{max}$ (A) = 70 volt

$70 = 70 \sin (100\pi t + 0{,}380)$

$\sin (100\pi t + 0{,}380) = 1$

$100\pi t + 0{,}380 = \frac{1}{2}\pi$ rad

$100\pi t + 0{,}380 = 1$

$100\pi t = 1{,}1908$

$t = 0{,}037$ det $= 37{,}4$ ms

c

